

Clause C1: $p \vee \neg q$

Clause C2: $p \rightarrow \neg p \equiv \neg p \vee \neg p$ (By conditional-disjunction equivalence)

Clause C3: $\neg q \rightarrow (\neg p \vee \neg p) \equiv \neg(\neg q) \vee (\neg p \vee \neg p)$ (Conditional-disjunction equivalence) $\equiv q \vee (\neg p \vee \neg p)$ (Double-negation Law)

By Resolution on C1, C2 we get C4: $\neg q \vee \neg p$

By Resolution on C4, C3 we get C6: $(\neg p \vee \neg p) \vee \neg p$

A contradiction cannot be reached.

$\therefore$ The conclusion is not valid

35) Determine whether this argument, taken from Kalish and Montague [KaMo64], is valid.

If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. Therefore, Superman does not exist.

Solution:

- p : Superman is able to prevent evil
- q : Superman is willing to prevent evil
- r : Superman prevents evil
- s : Superman is impotent
- t : Superman is malevolent
- e : Superman exists

Premise:

- 1. $(p \wedge q) \rightarrow r$
- 2. $\neg p \rightarrow s$
- 3. $\neg q \rightarrow t$
- 4. $\neg r$
- 5. $e \rightarrow (\neg s \wedge \neg t)$

Conclusion: $\neg e$